

1. What is C language?

C is a general-purpose, procedural, and imperative programming language developed in the 1970s by Dennis Ritchie at Bell Labs. It's one of the most popular and influential programming language of all time.

It's considered a "mid-level" language because it offers low-level access to memory, allowing for efficient and fast code, but also has the structure of high-level languages.

2. Applications of C-programming.

- => C is used to develop core components of operating systems such as windows, linux and macos.
- => C is applied in embedded systems in small devices such as washing machines, microwave ovens and printers.
- => C is used to create game engines.
- => C is applied to develop database engines. MySQL database is implemented using C.
- => C is employed to create programs for devices and sensors of IoT.
- => C is employed to create desktop applications.
Eg:- notepad++.

3. What is Variable?

=> Variable act like as a containers, which is used to store the information.

=> ~~Info~~ Datatype decides what type of information holds in the variable.

4. What are different data types in C programming?

In C programming, data types determine the type of data a variable can hold, the operations that can be performed on it, and the amount of memory allocated to store it.

=> There are four types of data types. They are.

Basic data type :- int, char, float, double.

Derived data type :- array, pointer, structure, union.

Enumeration data type :- enum.

void data type :- void.

5. What is format Specifier?

Format specifiers tell what type of information holds in the variable.

Purpose:

=> Control output format: In `printf()`, format specifiers define how variables are displayed.

=> Specify input format: In `scanf()`, format specifiers define how to interpret input data.